Higgs Boson at the LHC - Who produced this Higgs?

Master's Degree in Physics

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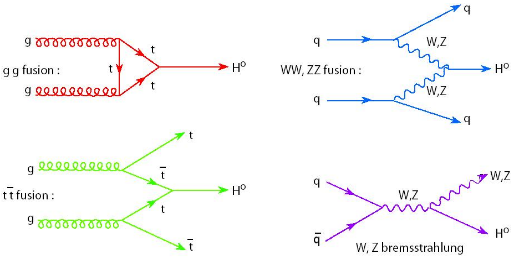
DEGLI STUDI
DI MILANO

Higgs Boson Production Mechanisms

Cross-sections at $\sqrt{s} = 8$ TeV for $m_H = 125$ GeV

Production	σ_{H} (pb)	(%)
ggF	19.27	87.2%
VBF	1.578	7.1%
WH	0.7046	3.2%
ZH	0.4153	1.9%
$t\bar{t}H$	0.1293	0.6%

Source: Prof. Fanti's Slides



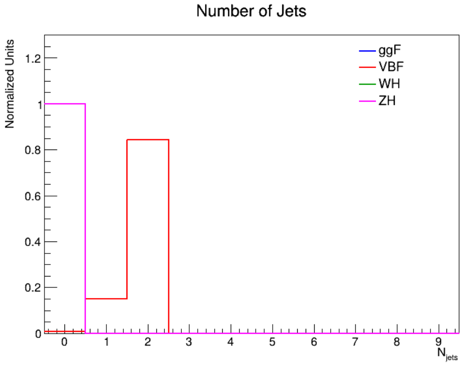
Source: sites.uci.edu

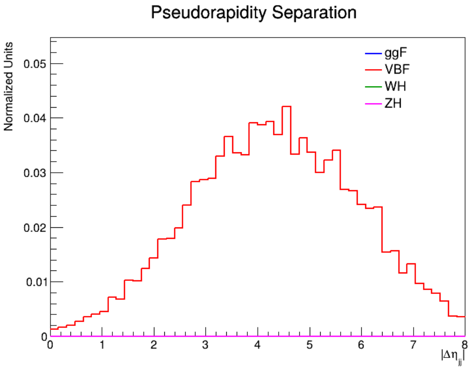
Simulation Workflow: Monte Carlo Generation

Hard Process	`MadGraph5_aMC@NLO`
Parton Level (LHE)	*Validation & Kinematics*
Shower & Hadronization	`Pythia8`
Detector Simulation	`Delphes`

- **Assumptions:** Inclusive Higgs decays (SM BRs) and `Delphes` fast-simulation.
- **Analysis Caveat:** Discrepancies in decay phase space and detector modeling are qualitatively considered during sample comparisons.

Simulation Workflow: Parton-Level Distributions





Sanity check of the parton-level kinematics: the distributions of the **number of jets** and the **pseudorapidity gap between the two leading jets** are consistent with theoretical expectations for each production mechanism.

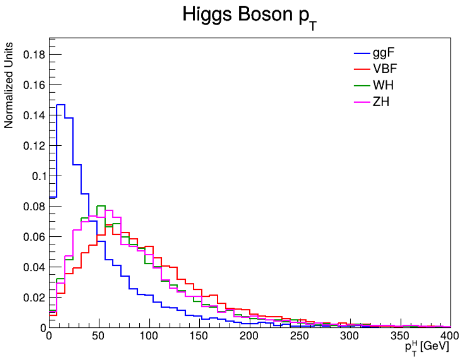
Experimental Signatures at Detector Level

- Following the Delphes simulation, the analysis relies exclusively on reconstructed kinematic variables to mimic experimental conditions.
- **Signal Region (SR) Strategy:** Selection of a single, highly-discriminating variable for each mechanism to optimize the signal-to-background ratio.
- **Golden Variables Identified:**

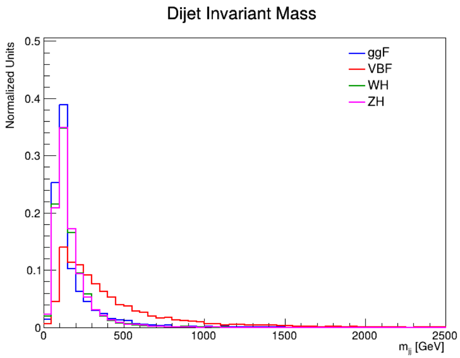
Process	Key Variable	SR Window
ggF	p_T^H	$[0.0, 100.0]$ GeV
VBF	m_{jj}	$[400.0, 1100.0]$ GeV
WH	m_T^W	$[0.0, 120.0]$ GeV
ZH	m_{ll}	$[80.0, 100.0]$ GeV

Experimental Signatures at Detector Level - ggF & VBF

ggF Signature



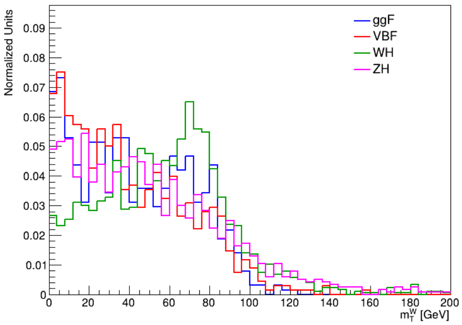
VBF Signature



Experimental Signatures at Detector Level - WH & ZH

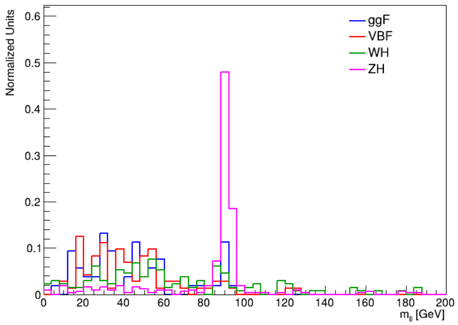
WH Signature

W Transverse Mass



ZH Signature

Dilepton Invariant Mass



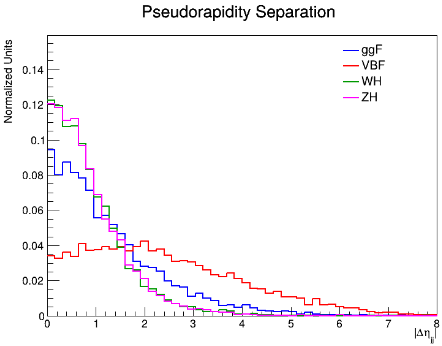
Deep Dive: VBF Topology & Backgrounds

Background Discrimination:

- **Irreducible (ggF + ≥ 2 jets):** Shares the identical $H + jj$ final state. Separable *only* by exploiting the unique VBF kinematics.
- **Reducible (WH / ZH):** Associated vector bosons. Suppressed via m_T^W and m_{ll} shape.

Key VBF Signatures:

- m_{jj}: Hard invariant mass spectrum.
- $\Delta\eta_{jj}$: Large pseudorapidity gap due to forward-backward tagging jets.



Blind Samples Analysis

1. Reconstruction

$H \to ZZ^* \to 4\ell$ candidate

$\Downarrow$

2. Mass Window

Signal isolation ($m_{4\ell} \simeq 125$ GeV)

$\Downarrow$

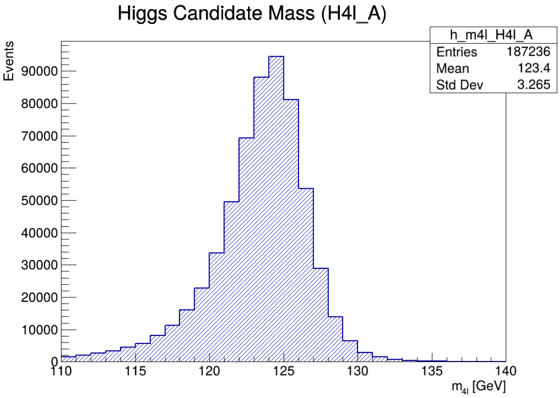
3. Kinematics Extraction

Associated features (jets, $l^{\pm}$, E_T^{miss})

$\Downarrow$

4. Shape Matching

Data vs. MadGraph MC



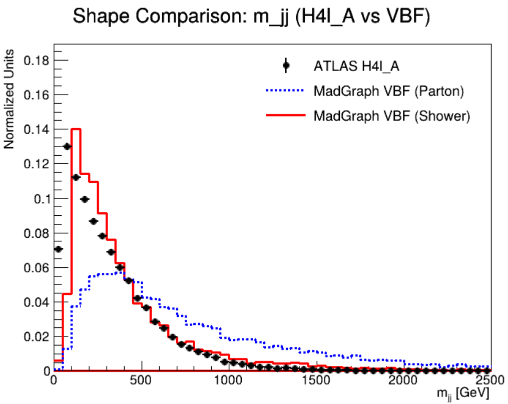
VBF Identification

VBF Signal Region Definition:

- **Phase Space Selection:** Decided on the dijet system constructed from the two leading-p_T jets.
- **Inferred SR Cut:** $m_{jj} \in [400, 1100]$ GeV to maximize VBF sensitivity and reject QCD-dominated background shapes.

VBF Matching (Sample A):

- The reconstructed m_{jj} distribution of **Sample A** exhibits excellent shape compatibility with the VBF MC template.

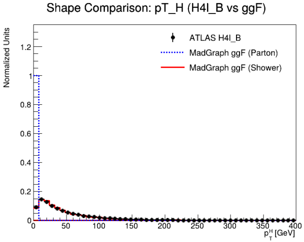


Events in SR: Data = 27.8% || MC expectation = 29.9%

Kinematic Matching for ggF, ZH, and WH

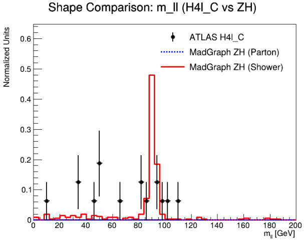
Identifying the remaining blind samples:

ggF (Sample B)



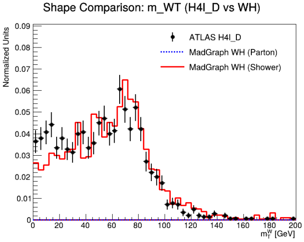
Events in SR:
Data = 86.7% || MC = 88.0%

ZH (Sample C)



Events in SR:
Data = 43.8% || MC = 77.7%

WH (Sample D)



Events in SR:
Data = 98.5% || MC = 97.2%

Each sample exhibits strong shape compatibility with its respective MC expectation within the specifically defined Signal Region.

Statistical Validation (χ^2 Shape Test)

To quantitatively confirm the visual matches, a χ^2 shape test was performed comparing the Data samples to the MC templates in the corresponding Signal Regions.

Blind Sample	Best MC Match	Key Variable	Shape Test (χ^2/NDF)
Sample A	VBF	m_{jj}	$\mathbf{7.08}$
Sample B	ggF	p_T^H	$\mathbf{3.03}$
Sample C	ZH	$m_{\ell\ell}$	$\mathbf{5.06}$
Sample D	WH	m_T^W	$\mathbf{2.19}$

- The shape tests strongly reject the null hypotheses for mismatched combinations (e.g., Sample D vs ZH yields $\chi^2/\text{NDF} > 999$).
- The designed kinematic discriminants successfully and reliably decouple the Higgs production mechanisms at the detector level.

Conclusions & Limitations

Successes & Key Results

- ✓ **Kinematic Profiling:** Identified optimal "golden variables" (m_{jj}, p_T^H, $m_{\ell\ell}$, m_T^W) to successfully isolate each Higgs production mode.
- ✓ **Sample Unblinding:** Correctly matched all anonymous ATLAS Open Data samples (A = VBF, B = ggF, C = ZH, D = WH).

Current Limitations & Future Outlook

- ! **1D Discrimination:** Quantitative matching relied strictly on single variables, ignoring multivariate kinematic correlations.
- ! **Decay Decoupling (Outlook):** Future steps require forcing $H \to 4\ell$ at the generator level to cleanly subtract the Higgs products and purely study the associated signature (jets, extra $l^{\pm}$, E_T^{miss}).

Higgs Boson at the LHC - Who produced this Higgs?

Thank you for listening!

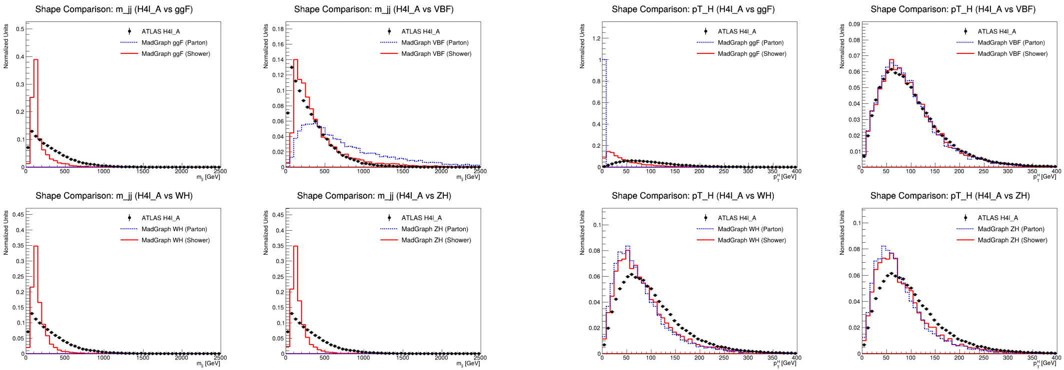
Any questions?

BACKUP

Backup: Extended Kinematic Agreement (VBF)

Sample A vs VBF Expectation

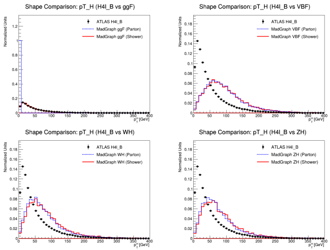
Beyond the key signal region variable (m_{jj}), the full topological profile of the sample is in excellent agreement with the Vector Boson Fusion hypothesis.



Backup: Hypothesis Rejection for ggF, ZH, WH

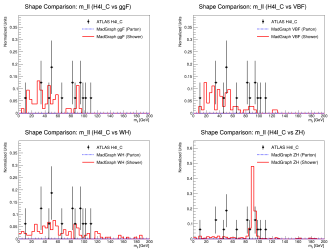
Cross-checking all hypotheses in their respective Signal Regions:

Sample B vs All



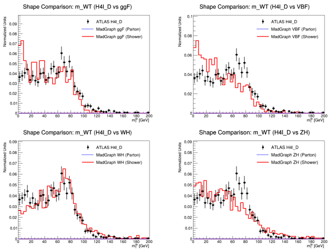
Key: p_T^H. Strong match for **ggF**.

Sample C vs All



Key: $m_{\ell\ell}$. Strong match for **ZH**.

Sample D vs All

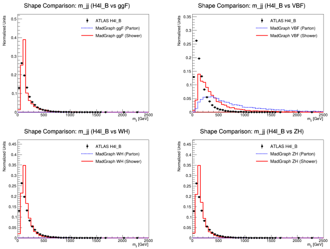


Key: m_T^W. Strong match for **WH**.

The grid comparisons demonstrate that each sample unambiguously prefers a single production mechanism, with alternative hypotheses yielding χ^2/NDF values orders of magnitude larger.

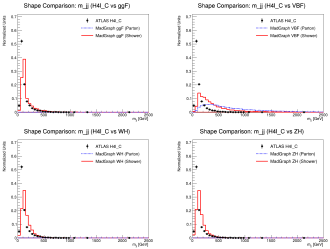
Backup: VBF Signature (m_{jj}) on Non-VBF Samples

Sample B (ggF)



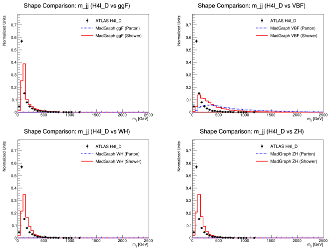
Severe shape mismatch in SR.

Sample C (ZH)



Severe shape mismatch in SR.

Sample D (WH)

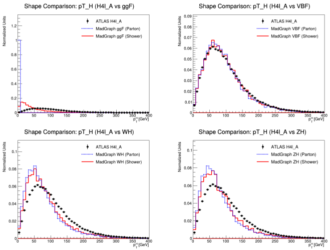


Severe shape mismatch in SR.

Applying the VBF signal region selection ($m_{jj} \in [400, 1100]$ GeV) to non-VBF samples yields drastically poor fits, proving the high rejecting power of the dijet invariant mass against other mechanisms.

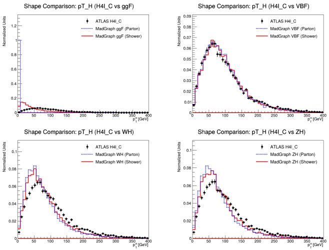
Backup: ggF Signature (p_T^H) on Non-ggF Samples

Sample A (VBF)



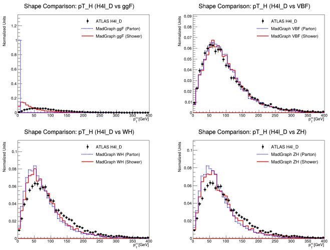
Poor fit across all templates.

Sample C (ZH)



Poor fit across all templates.

Sample D (WH)

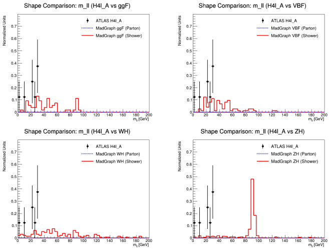


Poor fit across all templates.

The low-p_T^H signal region correctly isolates ggF; when evaluating samples with hard scatterings (VBF, VH) in this regime, the observed kinematics are entirely incompatible with the expected shapes.

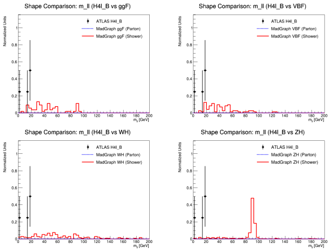
Backup: ZH Signature ($m_{\ell\ell}$) on Non-ZH Samples

Sample A (VBF)



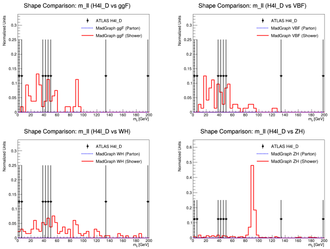
Lack of a clear Z peak.

Sample B (ggF)



Lack of a clear Z peak.

Sample D (WH)

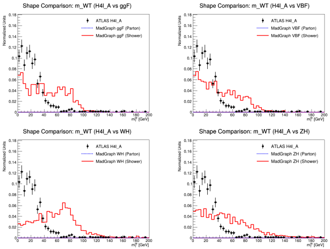


Lacks the extra lepton pair.

The invariant mass of the extra leptons $m_{\ell\ell}$ searches for an associated Z boson. Non-ZH samples fail to populate the expected Z-mass window ($[80, 100]$ GeV), making it an unambiguous discriminant.

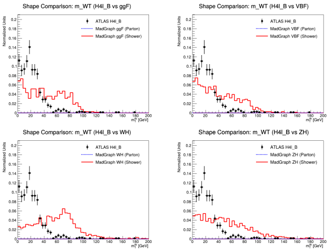
Backup: WH Signature (m_T^W) on Non-WH Samples

Sample A (VBF)



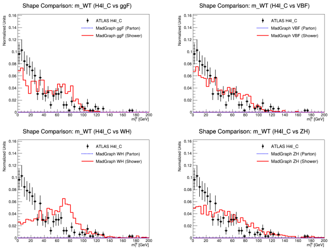
Inconsistent transverse mass.

Sample B (ggF)



Inconsistent transverse mass.

Sample C (ZH)



Inconsistent transverse mass.

The transverse mass m_T^W computed from the extra lepton and E_T^{miss} acts as an excellent veto. Samples lacking an associated W boson production display χ^2 values that completely reject the WH hypothesis.